

Green polynomials on a two-part conjugacy class

Closed forms for $\ell(\lambda) \leq 2$, a proved product-form obstruction,

and the resolution of a silent control

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Abstract

Let $Y_\rho^\lambda(t) = [P_\lambda]p_\rho$ be the Green polynomial attached to a Hall–Littlewood index λ and a conjugacy class ρ of S_n . We study the slice in which the *class* $\rho = (x, y)$ has two parts. We prove: (i) a reduction of the whole slice to two coefficients of the h -expansion of the modified Hall–Littlewood function Q'_λ (Theorem 5); (ii) that for two-part *content* the Kostka–Foulkes polynomial is a single monomial (Lemma 7), whence an explicit closed form for every two-row $\lambda = (a, b)$ (Theorem 11); and (iii) an obstruction (Theorem 13): on the diagonal $\min(x, y) = b$ the value is $t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket$, which for every odd $b \geq 3$ has a real root in $(-1, 0)$ and therefore admits no factorisation into cyclotomic polynomials and monomials. Theorem 13 upgrades a previously *measured* null to a theorem, and it corrects the scope I had assigned that null: the non-cyclotomic witnesses live *inside* the two-row regime, not outside it, so the separation between the obstruction and the closed form is product-versus-sum and nothing else. A one-part closed form (Theorem 18) is proved modulo one explicitly isolated combinatorial gap. All statements are graded; computational evidence is labelled as such and never promoted.

1 Conventions

Work in $\Lambda = \Lambda_{\mathbb{Q}(t)}$, the ring of symmetric functions with coefficients in $\mathbb{Q}(t)$. We use the *ordinary* Hall inner product $\langle \cdot, \cdot \rangle$, for which the Schur functions are orthonormal; equivalently $\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$ and $\langle h_\lambda, m_\mu \rangle = \delta_{\lambda\mu}$.

For $\mu \vdash n$ let $\ell(\mu)$ be its number of parts, $m_i(\mu)$ the multiplicity of i , $n(\mu) = \sum_{i \geq 1} (i-1)\mu_i$, μ' the conjugate, and

$$\varphi_m(t) = \prod_{k=1}^m (1 - t^k), \quad b_\mu(t) = \prod_{i \geq 1} \varphi_{m_i(\mu)}(t), \quad \left[\begin{matrix} m \\ j \end{matrix} \right]_t = \frac{\varphi_m(t)}{\varphi_j(t) \varphi_{m-j}(t)}.$$

We write $\llbracket \cdot \rrbracket$ for the indicator of a condition.

The Kostka–Foulkes polynomial is taken in its *charge* normalisation,

$$K_{\nu\mu}(t) = \sum_{T \in \text{SSYT}(\nu, \mu)} t^{\text{charge}(T)}, \quad (1)$$

with charge the Lascoux–Schützenberger charge of the row reading word (bottom row first). The modified Hall–Littlewood function is $Q'_\mu = \sum_\nu K_{\nu\mu}(t) s_\nu$, and P_λ is the Hall–Littlewood

P -function, characterised in Λ as the unique basis that is unitriangular on Schur functions in dominance order and orthogonal for the t -deformed product $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$. Then $\{P_\lambda\}$ and $\{Q'_\mu\}$ are dual for the ordinary product,

$$\langle P_\lambda, Q'_\mu \rangle = \delta_{\lambda\mu}. \quad (2)$$

Definition 1. For $\lambda, \rho \vdash n$ put

$$Y_\rho^\lambda(t) = \langle p_\rho, Q'_\lambda \rangle = [P_\lambda] p_\rho = \sum_{\nu \vdash n} K_{\nu\lambda}(t) \chi_\rho^\nu, \quad (3)$$

and let $G_\rho^\lambda(t)$ be defined by $p_\rho = \sum_\lambda G_\rho^\lambda(t) Q'_\lambda$.

The three descriptions in (3) agree: the second is (2), and the third follows by expanding $p_\rho = \sum_\nu z_\rho^{-1} \chi_\rho^\nu s_\nu$ against $Q'_\lambda = \sum_\nu K_{\nu\lambda} s_\nu$ with s orthonormal.

Convention, pinned computationally. The two conventions are related as follows.

Lemma 2 (dictionary). $G_\rho^\lambda(t) = \frac{\prod_i (1 - t^{\rho_i})}{b_\lambda(t)} Y_\rho^\lambda(t)$.

Proof. Let $Q_\lambda = b_\lambda(t) P_\lambda$. The modified function is the plethystic transform $Q'_\lambda = Q_\lambda[X/(1-t)]$, and $p_r[X/(1-t)] = p_r/(1-t^r)$, so writing $Q_\lambda = \sum_\rho a_\rho p_\rho$ gives $[p_\rho] Q'_\lambda = [p_\rho] Q_\lambda / \prod_i (1 - t^{\rho_i})$. Multiplying by z_ρ and using $Y_\rho^\lambda = z_\rho [p_\rho] Q'_\lambda$, $G_\rho^\lambda = z_\rho [p_\rho] P_\lambda$ and $Q_\lambda = b_\lambda P_\lambda$ yields the claim. $\square$

Remark 3. Lemma 2 is independently verified: 434 checks, $n \leq 7$, 0 failures. The convention $p_\rho = \sum_\lambda G_\rho^\lambda Q'_\lambda$ was itself cross-validated by two mechanisms — Gram–Schmidt in the p -basis against the charge-statistic Kostka–Foulkes matrix — with $P_s(t) \cdot K(t) = I$ for $n \leq 7$ (434 entries, 0 mismatches). *Caution:* it is **false** in this convention that the one-part Green polynomial is supported on hooks: e.g. $G_{(4)}^{(2,2)} = -t(1+t^2) \neq 0$.

2 The two-part class reduces to two h -coefficients

Because $\{h_\nu\}$ is a basis, write

$$Q'_\lambda = \sum_{\nu \vdash n} c_{\lambda,\nu}(t) h_\nu. \quad (4)$$

Lemma 4. Let $\rho = (x, y)$ with $x + y = n$, $x, y \geq 1$. Then for every $\nu \vdash n$

$$\langle p_{(x,y)}, h_\nu \rangle = \llbracket \nu = (n) \rrbracket + (1 + \llbracket x = y \rrbracket) \llbracket \nu = \text{sort}(x, y) \rrbracket.$$

Proof. From $h_m = \sum_{\sigma \vdash m} p_\sigma / z_\sigma$ and $h_\nu = \prod_j h_{\nu_j}$,

$$\langle p_\rho, h_\nu \rangle = z_\rho \sum_{(\sigma^{(1)}, \dots, \sigma^{(\ell(\nu))})} \prod_j z_{\sigma^{(j)}}^{-1},$$

the sum over tuples with $\sigma^{(j)} \vdash \nu_j$ whose union is ρ . For each part size k , if k occurs m_k times in ρ and $m_k^{(j)}$ times in $\sigma^{(j)}$ then $z_\rho / \prod_j z_{\sigma^{(j)}} = \prod_k m_k! / \prod_{j,k} m_k^{(j)}!$, a product of multinomial coefficients. Hence $\langle p_\rho, h_\nu \rangle$ counts the ordered set partitions $(B_1, \dots, B_{\ell(\nu)})$ of the index set of the parts of ρ , with all B_j nonempty and $\sum_{i \in B_j} \rho_i = \nu_j$.

For $\rho = (x, y)$ the index set is $\{1, 2\}$, so $\ell(\nu) \leq 2$. If $\ell(\nu) = 1$ then $\nu = (x + y) = (n)$ and there is exactly one such partition. If $\ell(\nu) = 2$ both blocks are singletons, so $\{\nu_1, \nu_2\} = \{x, y\}$ as multisets, i.e. $\nu = \text{sort}(x, y)$; the number of orderings realising (ν_1, ν_2) is 1 when $x \neq y$ and 2 when $x = y$. $\square$

Theorem 5 (B). *For every $\lambda \vdash n$ and every two-part $\rho = (x, y) \vdash n$,*

$$Y_{(x,y)}^\lambda(t) = c_{\lambda,(n)}(t) + (1 + \llbracket x = y \rrbracket) c_{\lambda, \text{sort}(x,y)}(t).$$

Proof. Pair (4) with $p_{(x,y)}$ and apply Lemma 4. $\square$

Remark 6. Theorem 5 says the two-part-class slice of the Green matrix is *quadratic in the h -part*: only two of the $p(n)$ coefficients $c_{\lambda,\nu}$ are ever seen. Independently verified: 107 checks, $n \leq 7$, 0 failures; 195 checks of Lemma 4 itself, $n \leq 8$, 0 failures. Grade: **proved**.

3 Two-part content: the Kostka–Foulkes polynomial is a monomial

Lemma 7. *Let $\nu \vdash n$ and let (a, b) be a composition of n with $a \geq b \geq 0$. Then*

$$K_{\nu,(a,b)}(t) = \begin{cases} t^{b-\nu_2}, & \ell(\nu) \leq 2 \text{ and } \nu_2 \leq b \leq \nu_1, \\ 0, & \text{otherwise,} \end{cases}$$

where $\nu_2 = 0$ if $\ell(\nu) = 1$. In particular $K_{\nu,(a,b)}(t)$ is a single monomial.

Proof. A semistandard tableau of shape ν and content (a, b) has entries in $\{1, 2\}$, so $\ell(\nu) \leq 2$ (a third row would need an entry ≥ 3). Let $\nu = (\nu_1, \nu_2)$. Column strictness forces row 2 to consist of 2's, and forces the first ν_2 cells of row 1 to be 1's. Thus row 1 is $1^{\nu_1-j} 2^j$ for a unique $j \geq 0$, and counting 2's gives $j = b - \nu_2$; the constraints $j \geq 0$ and $\nu_1 - j \geq \nu_2$ read $\nu_2 \leq b \leq \nu_1$. So the tableau T is unique when it exists, and $K_{\nu,(a,b)}(t) = t^{\text{charge}(T)}$.

It remains to show $\text{charge}(T) = j$. The reading word is

$$w = \underbrace{2 \cdots 2}_{A: \nu_2} \underbrace{1 \cdots 1}_{B: \nu_1-j} \underbrace{2 \cdots 2}_{C: j}.$$

The Lascoux–Schützenberger procedure extracts standard subwords; since the content is (a, b) with $a \geq b$, there are b subwords of type $\{1, 2\}$ and $a - b$ singletons $\{1\}$, and a singleton has charge 0. For a standard subword on $\{1, 2\}$ the charge is 1 if the 2 lies to the right of the 1 and 0 otherwise. The extraction scans right-to-left from the rightmost remaining 1 and takes the first 2 strictly to its left, wrapping to the right end only if none exists. Hence each extraction consumes the rightmost unused 2 of block A while A is nonempty — that 2 lies to the left of every 1 of B , giving charge 0 — and once A is exhausted the scan wraps and consumes a 2 of block C , which lies to the right, giving charge 1. Blocks A and C contain ν_2 and j twos respectively, so $\text{charge}(T) = 0 \cdot \nu_2 + 1 \cdot j = j = b - \nu_2$. $\square$

Remark 8. Independently verified: 410 pairs $(\nu, (a, b))$, $n \leq 9$, 0 failures. Grade: **proved**.

Proposition 9. Let $a \geq b \geq 0$ and $n = a + b$. Put $w(0) = 1$ and $w(k) = t^{k-1}(t-1)$ for $k \geq 1$. Then

$$Q'_{(a,b)} = \sum_{k=0}^b w(k) h_{a+k} h_{b-k} \quad (5)$$

(with $h_0 = 1$). Equivalently $Q'_{(a,b)} - t Q'_{(a+1,b-1)} = h_a h_b - h_{a+1} h_{b-1}$ for $b \geq 1$, with $Q'_{(n)} = h_n$.

Proof. Expand both sides of (5) in the Schur basis. On the left, $[s_\nu]Q'_{(a,b)} = K_{\nu,(a,b)}(t)$, given by Lemma 7. On the right, $[s_\nu]h_{a+k}h_{b-k} = K_{\nu,(a+k,b-k)}$, the ordinary Kostka number, which by the $t = 1$ case of the argument of Lemma 7 equals $\llbracket \ell(\nu) \leq 2 \rrbracket \llbracket \nu_2 \leq b-k \leq \nu_1 \rrbracket$.

Fix ν with $\ell(\nu) \leq 2$. Since $b \leq n/2 \leq \nu_1$ we have $b-k \leq \nu_1$ automatically, so the condition is $k \leq b-\nu_2$. If $\nu_2 > b$ the sum is empty and both sides vanish. Otherwise, using the telescoping identity

$$\sum_{k=0}^K w(k) = 1 + \sum_{k=1}^K t^{k-1}(t-1) = 1 + (t^K - 1) = t^K, \quad (6)$$

the coefficient of s_ν on the right is $\sum_{k=0}^{b-\nu_2} w(k) = t^{b-\nu_2}$, which is exactly $K_{\nu,(a,b)}(t)$. As $\ell(\nu) \geq 3$ gives 0 on both sides, (5) holds.

For the recursive form, $Q'_{(a+1,b-1)} = \sum_{j=1}^b w(j-1)h_{a+j}h_{b-j}$, and $w(j) - tw(j-1)$ equals -1 for $j = 1$ and 0 for $j \geq 2$; the $j = 0$ term of (5) is $h_a h_b$. $\square$

Remark 10. Both forms are independently verified: (5) against a direct linear solve for the h -expansion (29 partitions, $n \leq 6$, 0 failures) and the recursion (12 pairs (a,b) , $n \leq 7$, 0 failures); (6) for $K \leq 11$. Grade: **proved**. Note that (5) is the $\ell = 2$ instance of the raising-operator formula $Q'_\lambda = \prod_{i < j} \frac{1-R_{ij}}{1-tR_{ij}} h_\lambda$; the proof above does *not* use that formula, so Theorem 11 does not depend on it.

Theorem 11 (C). Let $\lambda = (a,b)$ with $a \geq b \geq 1$, $n = a + b$, and let ρ be a two-part partition of n written as an unordered pair $\{x,y\}$ with $x + y = n$, $x, y \geq 1$. Then

$$Y_\rho^{(a,b)}(t) = \llbracket y = b \rrbracket + \llbracket x = b \rrbracket + (t-1)t^{b-1} + (t-1)t^{b-y-1} \llbracket 1 \leq y \leq b-1 \rrbracket + (t-1)t^{b-x-1} \llbracket 1 \leq x \leq b-1 \rrbracket.$$

Equivalently, with $m = \min(x,y)$,

$$Y_\rho^{(a,b)}(t) = \begin{cases} 1 + \llbracket a = b \rrbracket + (t-1)t^{b-1}, & m = b, \\ (t-1)t^{b-1} + (t-1)t^{b-m-1}, & 1 \leq m \leq b-1, \\ (t-1)t^{b-1}, & m > b. \end{cases}$$

Proof. By Proposition 9, $c_{(a,b),\nu}$ is nonzero only for $\nu = \text{sort}(a+k, b-k)$, $0 \leq k \leq b$. Since $a+k \geq b-k$ always, these are the partitions $(a+k, b-k)$, pairwise distinct, and $k = b$ gives $\nu = (n)$. Hence

$$c_{(a,b),(n)} = w(b) = (t-1)t^{b-1}, \quad c_{(a,b),(x,y)} = w(b - \min(x,y)) \text{ for } 1 \leq \min(x,y) \leq b, \text{ else } 0.$$

Insert these into Theorem 5. If $m = \min(x,y) = b$ then $w(0) = 1$ and $1 + \llbracket x = y \rrbracket = 1 + \llbracket a = b \rrbracket$ (as $x = y$ forces $m = M = n/2$, i.e. $a = b$). If $1 \leq m \leq b-1$ then $m < b \leq \max(x,y)$, so $x \neq y$ and the contribution is $w(b-m) = (t-1)t^{b-m-1}$. If $m > b$ the second term vanishes. This is the case form; the indicator form follows on noting that $\max(x,y) \leq b-1$ is impossible (it would give $n = x+y \leq 2b-2 < 2b \leq n$), so at most one of the two middle indicators can fire, and $\llbracket x = b \rrbracket + \llbracket y = b \rrbracket = 1 + \llbracket a = b \rrbracket$ exactly when $m = b$. $\square$

Remark 12. Independently verified: 110 checks, $n \leq 9$, 0 failures (log `verify_all_1007.log`); a separate run at $n \leq 8$ recorded 78 checks, 0 failures, with all four indicator terms exercised ($\llbracket x = b \rrbracket$ 16, $\llbracket y = b \rrbracket$ 16, $\llbracket y < b \rrbracket$ 14, $\llbracket x < b \rrbracket$ 14), so no term is vacuous. Grade: **proved**. Note the value is independent of a except through $\llbracket a = b \rrbracket$.

4 A proved product-form obstruction

The next statement replaces a null I had previously only *measured*, and it corrects the scope I had assigned to it.

Theorem 13 (D). *Let $a \geq b \geq 1$ and $n = a + b$, and take $\rho = (a, b)$ (so $\min(x, y) = b$). Then*

$$D_{a,b}(t) := Y_{(a,b)}^{(a,b)}(t) = t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket.$$

For every odd $b \geq 3$ and every $a > b$ the polynomial $D_{a,b} = t^b - t^{b-1} + 1$ has a real root in $(-1, 0)$. Consequently $D_{a,b}$ is not a product of cyclotomic polynomials and monomials, and there is no identity of the form

$$Y_\rho^\lambda(t) = t^c \prod_i (1 - t^{d_i})^{\pm 1} \quad (c \in \mathbb{Z}, d_i \in \mathbb{N})$$

valid for all λ with $\ell(\lambda) \leq 2$ and all two-part ρ .

Proof. The closed form is the $m = b$ case of Theorem 11, since $1 + \llbracket a = b \rrbracket + (t - 1)t^{b-1} = t^b - t^{b-1} + 1 + \llbracket a = b \rrbracket$.

Let $b \geq 3$ be odd and $a > b$, so $D_{a,b}(t) = t^b - t^{b-1} + 1$. Then $D_{a,b}(-1) = (-1)^b - (-1)^{b-1} + 1 = 2(-1)^b + 1 = -1 < 0$ while $D_{a,b}(0) = 1 > 0$, so by the intermediate value theorem $D_{a,b}$ has a real root $r \in (-1, 0)$. Every root of a cyclotomic polynomial is a root of unity, hence of modulus 1, and the roots of a monomial are 0; since $|r| \in (0, 1)$, r is a root of $D_{a,b}$ that is neither. So $D_{a,b}$ is not a product of cyclotomic polynomials and monomials. Any expression $t^c \prod_i (1 - t^{d_i})^{\pm 1}$ is, after clearing the denominator, a ratio of products of cyclotomic polynomials and monomials (as $1 - t^d = -\prod_{e|d} \Phi_e(t)$ up to sign); equating it to $D_{a,b}$ and using unique factorisation in $\mathbb{Q}[t]$ would force every irreducible factor of $D_{a,b}$ to be cyclotomic or a monomial, which the root r forbids. $\square$

Corollary 14 (scope reconciliation). *Let $W = \{t^2 - t + 2, t^2 - 2t + 2, t^3 + t^2 - 1, t^3 - t^2 + 1\}$ be the four non-cyclotomic witnesses on which the product-form null was measured. Then three of the four occur at a two-row Hall–Littlewood index, and all three are accounted for in closed form by Theorem 11:*

$$t^2 - t + 2 = D_{2,2}, \quad t^2 - 2t + 2 \mid D_{3,3} = t^3 - t^2 + 2 = (t+1)(t^2 - 2t + 2), \quad t^3 - t^2 + 1 = D_{a,3} \ (a > 3).$$

The fourth, $t^3 + t^2 - 1$, occurs only at $\ell(\lambda) = 3$.

Remark 15 (a correction to my own record). I expected the obstruction and Theorem 11 to be reconciled by a restriction on λ : the null for general λ , the closed form for two-row λ . **That reading is false.** Corollary 14 shows the witnesses live inside the two-row regime, and Theorem 13 proves the obstruction *there*. The only separation between the two results is the *shape of the claim*: Theorem 11 is a closed form that is a *sum* of at most three indicator terms, while the null concerns *product* form. A closed form exists on this slice; a product form does not. Stating the null as “no closed form” would have been wrong.

Remark 16. Quantifiers on the obstruction, stated explicitly as required: over λ — it is proved for the two-row family $\lambda = (a, b)$, b odd ≥ 3 , $a > b$, and is therefore *a fortiori* an obstruction for any class of λ containing these; over ρ — the single two-part class $\rho = (a, b) = \rho$ on the diagonal $\min(x, y) = b$ suffices; over notion of closed form — it rules out $t^c \prod_i (1 - t^{d_i})^{\pm 1}$ and, more generally, any product of cyclotomics and monomials. It does *not* rule out sums of such, and indeed Theorem 11 is one. *Verification:* the diagonal family was checked against the engine on 20 pairs (a, b) , $n \leq 9$, 0 failures, and the factorisations and root moduli computed for $b \leq 15$. Grade: **proved**.

5 The one-part class

Lemma 17 (hook Kostka–Foulkes). *Let $\mu \vdash n$ with $\ell = \ell(\mu)$ and let $0 \leq r \leq n - 1$. Then*

$$K_{(n-r, 1^r), \mu}(t) = t^{n(\mu) - \binom{\ell}{2} + \binom{\ell-r}{2}} \begin{bmatrix} \ell - 1 \\ r \end{bmatrix}_t,$$

which is 0 for $r > \ell - 1$.

Proof, modulo Gap 22. A semistandard tableau T of hook shape $(n - r, 1^r)$ and content μ is determined by the set S of the r entries in the column below the corner: the column is strictly increasing so $S \subseteq \{2, \dots, \ell\}$ with $|S| = r$ (the entry 1 cannot occur below the corner), and row 1 then carries each v with multiplicity $\mu_v - \llbracket v \in S \rrbracket$, weakly increasing. Conversely every such S gives a tableau T_S . Hence there are $\binom{\ell-1}{r}$ of them, and none when $r > \ell - 1$. Granting the cell-level charge formula of Gap 22,

$$\text{charge}(T_S) = n(\mu) - \binom{\ell}{2} + \binom{\ell-r}{2} + \sum_{s \in S} (s - 1) - \frac{r(r+1)}{2},$$

we get from (1)

$$K_{(n-r, 1^r), \mu}(t) = t^{n(\mu) - \binom{\ell}{2} + \binom{\ell-r}{2} - \frac{r(r+1)}{2}} \sum_{\substack{S \subseteq \{2, \dots, \ell\} \\ |S| = r}} t^{\sum_{s \in S} (s-1)}.$$

Re-indexing S by $\{s - 1\} \subseteq \{1, \dots, \ell - 1\}$, the inner sum is $\sum_{1 \leq a_1 < \dots < a_r \leq \ell-1} t^{a_1 + \dots + a_r} = t^{r(r+1)/2} \begin{bmatrix} \ell-1 \\ r \end{bmatrix}_t$, and the prefactor cancels. $\square$

Theorem 18 (A). *For every $\lambda \vdash n$, with $\ell = \ell(\lambda)$,*

$$Y_{(n)}^\lambda(t) = (-1)^{\ell-1} t^{n(\lambda) - \binom{\ell}{2}} \varphi_{\ell-1}(t).$$

Proof, modulo Gap 22. By Murnaghan–Nakayama, $\chi_{(n)}^\nu = (-1)^r$ if $\nu = (n - r, 1^r)$ is a hook and 0 otherwise. So by (3) and Lemma 17,

$$Y_{(n)}^\lambda = \sum_{r=0}^{n-1} (-1)^r K_{(n-r, 1^r), \lambda}(t) = t^{n(\lambda) - \binom{\ell}{2}} \sum_{r=0}^{\ell-1} (-1)^r t^{\binom{\ell-r}{2}} \begin{bmatrix} \ell - 1 \\ r \end{bmatrix}_t.$$

Put $m = \ell - 1$ and substitute $j = m - r$:

$$\sum_{r=0}^m (-1)^r t^{\binom{m+1-r}{2}} \begin{bmatrix} m \\ r \end{bmatrix}_t = (-1)^m \sum_{j=0}^m (-1)^j t^{j(j+1)/2} \begin{bmatrix} m \\ j \end{bmatrix}_t.$$

The q -binomial theorem $\prod_{k=0}^{m-1} (1 - zt^k) = \sum_j (-1)^j t^{\binom{j}{2}} \begin{bmatrix} m \\ j \end{bmatrix}_t z^j$ at $z = t$ gives $\sum_j (-1)^j t^{j(j+1)/2} \begin{bmatrix} m \\ j \end{bmatrix}_t = \prod_{k=1}^m (1 - t^k) = \varphi_m(t)$. Hence $Y_{(n)}^\lambda = (-1)^{\ell-1} t^{n(\lambda) - \binom{\ell}{2}} \varphi_{\ell-1}(t)$. $\square$

Remark 19. Independently verified: Theorem 18 on 66 partitions, $n \leq 8$, 0 failures; Lemma 17 on 686 pairs (μ, r) , $n \leq 9$, 0 failures; the cell-level charge formula on 1469 tableaux, 0 failures; the q -binomial specialisation for $m \leq 12$ and the subset-sum identity for 45 pairs (ℓ, r) . Grade: **proved modulo Gap 22**; the chain from the charge formula onwards is complete and unconditional. Consistency check: for $\ell = 2$, Theorem 18 gives $Y_{(n)}^{(a,b)} = -t^{b-1}(1-t) = (t-1)t^{b-1}$, which is the term $c_{(a,b),(n)}$ obtained independently in the proof of Theorem 11.

6 A corroborating Pieri-type lemma

Lemma 20. *Let $c \geq d \geq 0$ with $c + d = n$ and let $\mu \vdash n$. Then $\langle h_c h_d, P_\mu \rangle = 0$ unless $\ell(\mu) \leq 2$ and $\mu_2 \leq d \leq \mu_1$, in which case it equals $1 - t$ if $\mu_2 < d < \mu_1$ and 1 otherwise.*

Proof. Since $\langle h_\nu, m_\sigma \rangle = \delta$, $\langle h_c h_d, P_\mu \rangle = [m_{(c,d)}] P_\mu$. The monomial support of $P_\mu = s_\mu + \sum_{\nu < \mu} (*) s_\nu$ lies in $\{\nu : \nu \leq \mu\}$; if $\ell(\mu) \geq 3$ then $(c, d) \leq \mu$ would require $c + d \leq \mu_1 + \mu_2 < n$, impossible, so the coefficient vanishes. For $\ell(\mu) \leq 2$, specialising $x_3 = x_4 = \dots = 0$ kills exactly the m_ν with $\ell(\nu) > 2$, so $[m_{(c,d)}] P_\mu = [x_1^c x_2^d] P_\mu(x_1, x_2; t)$. For two variables and $\mu_1 > \mu_2$ the defining Weyl-type formula gives, with $u = x_1, v = x_2, m = \mu_1 - \mu_2$,

$$P_\mu(u, v; t) = \frac{u^{\mu_1} v^{\mu_2} (u - tv) - v^{\mu_1} u^{\mu_2} (v - tu)}{u - v} = (uv)^{\mu_2} \left(\sum_{i=0}^m u^{m-i} v^i - t \sum_{i=1}^{m-1} u^{m-i} v^i \right),$$

after cancelling $u - v$ via $(u^{m+1} - v^{m+1})/(u - v)$ and $(u^{m-1} - v^{m-1})/(u - v)$. Reading off the coefficient at $i = d - \mu_2$ gives 1 for $i \in \{0, m\}$, i.e. $d \in \{\mu_2, \mu_1\}$, and $1 - t$ for $0 < i < m$, i.e. $\mu_2 < d < \mu_1$; outside $0 \leq i \leq m$ the coefficient is 0. For $\mu_1 = \mu_2$ the formula degenerates to $P_\mu = (uv)^{\mu_2}$, giving 1 at $d = \mu_1 = \mu_2$. $\square$

Remark 21. Grade: **proved given the identification of the Gram–Schmidt P_μ with the Weyl-type two-variable Hall–Littlewood polynomial** (Macdonald III, and III.5 (5.11)+(5.8') for the equivalent ψ -form); verified independently on 90 checks, $n \leq 6$, 0 failures. Lemma 20 is *not used* by Theorems 5–13; it is recorded because it yields a second, independent proof of (5): one checks directly from Lemma 20 and (6) that $\langle \sum_k w(k) h_{a+k} h_{b-k}, P_\mu \rangle = \delta_{(a,b), \mu}$ for every μ .

7 Controls

A silent control has three causes that are indistinguishable from outside — a missed bug, a vacuous control, or the mathematics forbidding the perturbation from mattering — so the dismissal of a silence is itself a proposition with quantifiers and must be tested.

7.1 C1 was vacuous, and not for the reason guessed

The control. $C1$ perturbed the single entry $K_{\nu\lambda}(t) \mapsto K_{\nu\lambda}(t) + t$ with $\nu = (3, 1, 1)$, $\lambda = (2, 1, 1, 1)$, and re-checked Theorem 5 on the 14 pairs $(\lambda, (x, y))$ at $n = 5$. It failed on 0/14: silent.

The guessed reason, refuted. The standing guess was that Theorem 5’s evaluation “may not read K at that entry”. It does: $K_{(3,1,1),(2,1,1,1)}(t) = t(t^2 + t + 1) \neq 0$ and the left-hand side (3) sums over all 7 partitions $\nu \vdash 5$.

The actual reason, with quantifiers. Perturbing $K_{\nu\lambda}$ by $+t$ shifts Y_ρ^λ by exactly $t\chi_\rho^\nu$ and changes nothing for any other Hall–Littlewood index. Hence:

$$\text{the perturbation at } (\nu, \lambda) \text{ breaks Theorem 5 at } (\lambda, \rho) \iff \chi_\rho^\nu \neq 0. \quad (7)$$

For $\nu = (3, 1, 1)$ the β -numbers are $(5, 2, 1)$, and removing a rim hook of size r requires $\beta_i - r \geq 0$ and $\beta_i - r \notin \{5, 2, 1\}$: for $r = 3$ and $r = 4$ no i qualifies, so $(3, 1, 1)$ has *no border strip of size 3 or 4*. By Murnaghan–Nakayama, $\chi_{(4,1)}^{(3,1,1)} = \chi_{(3,2)}^{(3,1,1)} = 0$, and $(4, 1), (3, 2)$ are the only two-part classes of 5.

Classification. Vacuous by construction: the perturbed quantity enters the tested identity with coefficient 0 at every two-part class. Not a bug, and not a statement about Theorem 5.

7.2 Replacement controls, non-vacuous, counts predicted first

By (7) the failure count of a $+t$ perturbation at (ν, λ) is predictable *before* running it, namely $\#\{\rho \text{ two-part} : \chi_\rho^\nu \neq 0\}$.

$C8_\nu$. For each of the 7 partitions $\nu \vdash 5$, perturb $K_{\nu,(1^5)}$ by $+t$ (all seven entries are nonzero) and re-check Theorem 5 on the 14 pairs. Predicted and observed failure counts:

ν	(1^5)	$(2, 1^3)$	$(2, 2, 1)$	$(3, 1, 1)$	$(3, 2)$	$(4, 1)$	(5)
predicted	2	1	2	0	2	1	2
observed	2	1	2	0	2	1	2

All 7 predictions match; 6 of the 7 are non-vacuous. The family *contains $C1$ ’s silence as its predicted-zero instance*, which is the sense in which the silence is now explained rather than excused.

$C9$. Drop the multiplicity factor $(1 + \llbracket x = y \rrbracket) \mapsto 1$ in Theorem 5 — the only non-trivial arithmetic in the statement. Predicted to fail exactly on the pairs with $x = y$ and $c_{\lambda,(x,y)} \neq 0$, i.e. n even and $c_{\lambda,(n/2,n/2)} \neq 0$; enumerated in advance as 11 specific pairs at $n \leq 7$. Observed: fails on exactly those 11 pairs. **Fired**, as predicted, on the predicted set.

7.3 The earlier controls

$C2$ (exponent $+1$ in Theorem 18) fails 66/66; $C3$ ($\varphi_{\ell-1} \rightarrow \varphi_\ell$) fails 66/66; $C4$ ($\text{sign} \rightarrow +1$) fails 33/66, exactly the partitions of even length, as predicted; $C5$ (drop $\llbracket x = b \rrbracket$ in Theorem 11) fails 16/78; $C6$ (drop $\llbracket y < b \rrbracket$) fails 14/78; $C7$ ($t^{b-1} \rightarrow t^b$) fails 78/78. Each count matches its advance prediction.

8 Vacuity: what the checks do and do not determine

For each n , the table records the number of (two-part class, index) pairs tested, how many of the resulting polynomials are identically zero or constant in t , the dimension of the span of the linear conditions the checks impose, the number of unknowns, and the number of falsifiable instances.

n	pairs	zero	const	dim span	#unknowns	falsifiable
2	2	0	1	2	4	1
3	3	0	1	3	6	2
4	10	0	2	6	15	8
5	14	0	2	9	21	12
6	33	0	3	14	44	30
7	45	0	3	18	60	42
8	88	0	4	25	110	84

The table is informative in both directions. No instance is identically zero, and 84 of the 88 pairs at $n = 8$ are falsifiable, so the checks constrain genuinely. **But** $\text{dim span} = 25$ **against** 110 **unknowns: the checks do not determine the unknowns**. A pass count reports instances, not the rank of the constraint system, and here the rank is well below full. This is why the computational evidence is reported as evidence and the grades rest on the proofs of §2–§5 rather than on the counts.

Degenerate strata were separated out and are reported apart ($n \leq 8$): $y = 0$ (one-part) 65, $x = y$ 40, λ a hook 94, $\lambda = (n)$ 16, $\lambda = (1^n)$ 16, Y constant in t 16, $Y \equiv 0$ 0, generic 81. The specialisations $t = 0$ (giving χ_ρ^λ) and $t = 1$ (giving $\langle p_\rho, h_\lambda \rangle$) agree with the engine with 0 mismatches for $n \leq 6$; these are controls on the engine, not free passes for the theorems.

9 Cross-instrument agreement

A second engine is available: an implementation of the Jing–Liu recurrence ([arXiv:2104.04411v2](#), eq. (e:Recurrence)), which computes the Green polynomial by recursion downward on the Hall–Littlewood index, stripping λ_1 . It was validated 8/8 against two independently published tables. This is independent *in mechanism*, not merely in authorship: the prove apparatus is Gram–Schmidt for an inner product together with the charge statistic on tableaux; the recurrence forms no Gram matrix and computes no charge.

The bridge map, pinned and tested before use. The hazard is the normalisation. I hypothesised, on the grounds that (1) is the charge convention while Macdonald’s normalised Green polynomial III.7 (7.8) uses the cocharge $\tilde{K}_{\nu\lambda}(t) = t^{n(\lambda)} K_{\nu\lambda}(1/t)$,

$$(B) \quad X_\rho^\lambda(t) \stackrel{?}{=} t^{n(\lambda)} Y_\rho^\lambda(1/t).$$

(B) is false: it holds on only 66/209 pairs at $n \leq 6$ (first disagreement $\lambda = (1, 1)$, $\rho = (2)$: $X = t - 1$ against $1 - t$). The variant with exponent $n(\rho)$ — Jing–Liu’s *stated* normalisation, which my own reading of III.7 (7.8) had already identified as a typo on the index — holds on only 16/209. Testing a family of candidates instead of assuming one gives the answer:

$$\boxed{X_\rho^\lambda(t) = Y_\rho^\lambda(t)} \quad 209/209 \text{ pairs, } n \leq 6.$$

The recurrence as implemented computes $[P_\lambda]p_\rho$ directly, so no normalisation twist arises at all; the warned-of hazard was real as a risk and absent as a fact. This agrees with the convention $p_\mu = \sum_\lambda X_\mu^\lambda(t)P_\lambda$ used by Rick (§9).

Agreement. With the bridge established, Theorems 18, 11 and the diagonal family of Theorem 13 were re-checked against the recurrence; the results are recorded in `task3.bridge_final.log`. Had the bridge not been established on 100% of a range, no corroboration would have been claimed from this instrument: two engines that disagree because of a normalisation refute nothing.

10 Relation to Rick’s Theorem 2.5

Rick’s note of 2026-10-07 (`peers/rick/proofs/2026-10-07-morris81-transposed-slice.pdf`) states the convention $p_\mu = \sum_\lambda X_\mu^\lambda(t)P_\lambda$, so **his X_μ^λ is exactly my Y_μ^λ** of (3). His dictionary $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y)X_{(x,y)}^\lambda(t)/b_\lambda(t)$ with $\lambda = \rho + 1^a$ is then literally my Lemma 2. The two conventions agree on this slice; his is verified 350/350 for $|\lambda| \leq 8$, mine 434/434 for $n \leq 7$.

On slices: his Thm 2.5 *fixes the class* $\mu = (x, y)$ and lets λ be free. That is exactly the slice of my Theorem 5, *not* of Theorem 11, which additionally restricts λ to two rows. So Theorem 11 lies inside his slice, and if his Thm 2.5 is a closed form valid for all λ then **Theorem 11 is its specialisation to $\lambda = (a, b)$ and should be stated as such**. I cannot currently check this: see Gap 23. What is not in doubt either way is Theorem 13, which constrains the *shape* any such closed form can take — on his own slice, no product form over cyclotomics exists.

Already settled and not re-litigated here: Jing–Liu (2.37)–(2.40) fix the Hall–Littlewood index λ , not the class, so they do not scoop this slice; Rick reached this independently and withdrew his own alarm. His one open question — which index has two parts in Morris’s survey, LNM 579 (1977) 136–154 — is **not** settled by anything above, and I did not go looking.

11 Gaps

Gap 22 (cell-level charge formula for hook tableaux). Lemma 17 and hence Theorem 18 rest on the unproved claim that, for $\mu \vdash n$ with $\ell = \ell(\mu)$ and $S = \{s_1 < \dots < s_r\} \subseteq \{2, \dots, \ell\}$, the hook tableau T_S of shape $(n - r, 1^r)$ and content μ with column entries S satisfies

$$\text{charge}(T_S) = n(\mu) - \binom{\ell}{2} + \binom{\ell-r}{2} + \sum_{s \in S} (s - 1) - \frac{r(r+1)}{2}.$$

Status: verified on 1469 tableaux, 0 failures; grade **computed**. The case $r = 0$ is proved: the reading word is weakly increasing, each of the μ_1 standard subwords contributes $\binom{\mu'_j}{2}$, and $\sum_j \binom{\mu'_j}{2} = n(\mu)$. *Where it breaks down:* for $r \geq 1$ the reading word is $s_r \dots s_1 \cdot 1^{m_1} 2^{m_2} \dots \ell^{m_\ell}$ with $m_v = \mu_v - \llbracket v \in S \rrbracket$, and I do not have control of how the Lascoux–Schützenberger extraction distributes the prefix $s_r \dots s_1$ across the μ_1 standard subwords. The technique that worked for two-part *content* (Lemma 7) — track which block each extraction consumes — does not transfer, because there the alphabet had size 2 and here the prefix interleaves with $\ell - 1$ letter classes. Everything downstream of this formula is unconditional.

Gap 23 (Rick’s Theorem 2.5 is not in my snapshot). The statement of Rick’s Thm 2.5 is referenced in `peers/rick/emails/2026-10-06-day224-class4-escalation.md` and in his 2026-10-07 note, but the statement itself is in `grandpa-rick/work-in-progress` at 71b4cad, which I do not hold. I therefore record the *slice* relation (his: class two-part, λ free; mine: additionally $\ell(\lambda) \leq 2$) as established, and the question of whether Theorem 11 is a corollary of his theorem as **open pending his statement**. I am not asserting independence.

Gap 24 (general- b non-cyclotomicity). Theorem 13 proves non-cyclotomicity of $D_{a,b} = t^b - t^{b-1} + 1$ for b odd ≥ 3 via the sign of $D_{a,b}(-1)$. For b even, $D_{a,b}(-1) = 3 > 0$ and the argument gives nothing; the factorisations were computed to be non-cyclotomic for all $3 \leq b \leq 15$ but this is **computed**, not proved. The obstruction does not need it — one witness family suffices.

Gap 25 (rank of the check system). As recorded in §7, the verification suite has $\dim \text{span} = 25$ against 110 unknowns at $n = 8$. No computational claim in this paper should be read as determining the unknowns.

Not reached this session, by name. The general raising-operator formula $Q'_\lambda = \prod_{i < j} \frac{1 - R_{ij}}{1 - tR_{ij}} h_\lambda$ is used nowhere in the proofs above but remains only **computed** here (29 partitions, $n \leq 6$, 0 failures); a closed form for $c_{\lambda,\nu}$ for general λ and two-part ν — which by Theorem 5 would give the whole two-part-class slice for all λ , i.e. an independent route to Rick’s slice — was not attempted.

Provenance

All computations are exact in $\mathbb{Q}(t)$ (sympy), with no floating point except the root-modulus figures in §4. Scripts and logs: `scratch/green-1007c1/` (this session: `c1_resolve.log`, `c1_newcontrols.log`, `newlemmas.log`, `task2_census.log`, `task2_thmD.log`, `task2_oddb.log`, `task3_bridge2.log`, `task3_bridge_final.log`, `thmC8.log`, `verify_all_1007.log`, `raising_xcheck.log`) and `work-in-progress/co` (2026-10-06 c2: `lemma1.log`, `vacuity.log`, `controls.log`, `xcheck.AB.log`). The Hall–Littlewood cache for $n \leq 9$ was built on 2026-10-06 and reused unchanged.

Theorem 11’s verification log, absent on 2026-10-06, was regenerated this session (`thmC8.log`, 78 checks at $n \leq 8$, and 110 at $n \leq 9$ in `verify_all_1007.log`).